

Real-Time SEEG Seizure Localization and Propagation Platform Proposal

Project Background

I propose developing a real-time SEEG analysis and visualization platform that identifies likely seizure-onset contacts, tracks how activity propagates across implanted electrodes, and displays these dynamics directly on a patient-specific 3D brain model.

During SEEG monitoring, clinicians may review hundreds of channels across multiple seizures to determine which contacts are involved earliest and distinguish seizure onset from rapid propagation. The goal of this platform is to organize these data into an intuitive, quantitative representation that supports clinical interpretation.

Objectives

The system would connect through a read-only stream to the existing SEEG recording system and provide:

- Live SEEG traces from implanted contacts.
- Contact-level onset scores estimating which electrodes show the earliest seizure-related changes.
- Animated seizure propagation across the patient's implanted electrode network.
- Patient-specific 3D visualization using preoperative MRI and postoperative CT to map electrode contacts onto the patient's anatomy.
- Signal quality and confidence measures so noisy channels and uncertain predictions are clearly identified.
- Seizure comparison, allowing multiple events from the same patient to be replayed and compared for consistency of onset and propagation.

The main clinical value is not simply automated seizure detection. It is helping distinguish where seizure activity appears to begin from where it subsequently propagates, while placing that information directly within the patient's anatomy.

This could make it easier to investigate:

- whether the same contacts consistently participate earliest across seizures
- whether strongly involved regions represent onset or rapid propagation
- how seizure activity travels through distributed cortical and subcortical networks
- how electrophysiological findings correspond to potential surgical or neuromodulation targets

The platform could also provide a shared visualization for multidisciplinary epilepsy discussions by allowing neurosurgeons, epileptologists, and researchers to replay the same seizure as both raw electrophysiology and an evolving anatomical network.

Methodology

1. Retrospective prototype:

Start with previously recorded SEEG seizures. For each event, calculate features such as changes in frequency power, line length, and temporal change points to estimate which contacts become involved first.

2. Anatomical integration:

Register postoperative CT to preoperative MRI, reconstruct electrode locations, and link each SEEG channel to its corresponding contact on a patient-specific 3D brain model.

3. Propagation visualization:

Estimate the recruitment time of each contact and animate the sequence of seizure spread through the implanted network.

4. Validation:

Compare system estimates with previously annotated seizures, including expert-identified onset contacts and recruitment patterns. Key measures would include agreement with clinical localization, recruitment-order accuracy, false alarms, and processing latency.

5. Real-time research deployment:

If retrospective results are promising, connect the same pipeline to a read-only live SEEG stream and initially run it in parallel with standard clinical monitoring as a research tool.

Long-Term Goal

The broader goal is to move from viewing a seizure primarily as hundreds of individual SEEG traces toward representing it as a patient-specific spatiotemporal brain network. We would be able to study where the seizure appears to begin, how confident are we in that localization, and how does it propagate through the patient's brain.